

Plant Monitor and Automated Watering System

Submitted by

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Problem Statement

Many people are always on the move from place to place due to busy work schedules, business demands, or frequent travel. Consequently, plant lovers often spend a couple of days away from their homes, leaving their indoor plants without any kind of real-time monitoring and care. Some sensitive plant species require specific soil moisture levels at different intervals depending on environmental factors, whereas others are highly susceptible to overwatering if irrigated incorrectly. Currently, maintaining healthy indoor greenery requires an individual to manually check soil conditions and attend to each plant pot independently from time to time. However, traditional manual plant care is highly prone to human error, often leading to either severe under-watering or excessive watering that causes root rot. All such environmental monitoring and irrigation control could be optimized and executed without the user necessarily being around or inside the home. Improper water management and neglected environmental conditions not only jeopardize plant health but can also encourage fatal mold growth, fungal diseases, or pest infestations. Therefore, we propose to design an Internet-of-Things-based smart plant monitoring and automated watering system, which will enable users to mitigate human error, prevent plant diseases, and remotely manage their plants' health from anywhere, anytime.

Abstract

This project presents the development of a Smart Plant Monitor and Automated Watering System designed specifically for indoor plants. The system uses an ESP 32 together with environmental sensors to monitor soil moisture, room temperature, and humidity in real time. A mini water pump controlled through a relay module automatically irrigates the plant whenever the soil moisture falls below a predefined threshold. The system aims to reduce human error in plant care and support users with busy schedules or frequent travel.

Objectives

1. Accurately measure room temperature and air humidity around the plant.
2. Monitor soil moisture levels using a capacitive soil moisture sensor.
3. Automatically activate a mini water pump when soil dryness exceeds the threshold.
4. Prevent overwatering by implementing intermittent pulse-watering logic.
5. Monitor environmental conditions to identify potential risks of plant diseases and pest development.

System Architecture

The system operates through a simple monitoring and irrigation process:

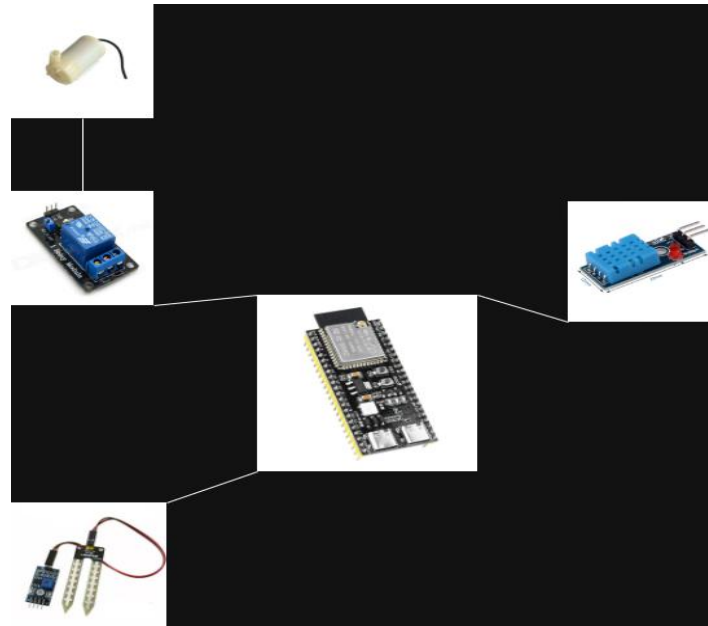
1. The Soil Moisture Sensor and DHT11 sensor collect environmental data every 5 minutes.
2. The ESP32 processes incoming sensor data and evaluates environmental conditions.
3. The system evaluates temperature, air humidity, and soil moisture conditions to identify environments that may encourage fungal diseases, mold growth, or pest infestations.
4. If the soil moisture level falls below approximately 35%–40% (corresponding to a sensor reading above 3500 ADC units), the relay module is activated automatically until the soil moisture reaches about 60% (approximately 2000-2500 ADC units)
5. The relay supplies power to the 5V mini water pump, which delivers water directly to the plant pot.
6. Once the soil moisture reaches the predefined optimal level, the pump is automatically turned off to prevent overwatering.

Hardware Implementation

4.1 Components List

Component	Quantity	Function
ESP32	1	Main microcontroller for processing sensor data
YL-69 Soil Moisture Sensor Module	1	Measures soil moisture
DHT11 Temperature and Humidity Sensor	1	Measures room temperature and humidity
5V 1-Channel Relay Module	1	Controls water pump activation
5V Mini Water Pump	1	Supplies water to the plant
1m Pipe (6 mm)	1	Transfers water to the pot
Breadboard MB-102	1	Prototype connections
Male-to-Female Jumper Wires	1	Electrical connections
Male -to-Male Jumper Wires	1	Electrical connections

4.2 Hardware Diagram



Step 1: Microcontroller Setup

- Place the ESP32 NodeMCU firmly onto the center of the MB-102 breadboard to establish a solid prototyping base.
- Identify the power rails on the breadboard, designating one row for VCC (5V/3.3V) and another for GND (Ground) connected to the ESP32 pins.

Step 2: Integrating the DHT11 Temperature and Humidity Sensor

- Connect the VCC pin of the DHT11 sensor to the 3.3V/5V power rail of the ESP32.
- Connect the GND pin of the DHT11 sensor to the common GND rail.
- Connect the Data (OUT) pin of the DHT11 to a designated Digital Input pin (e.g., GPIO 16 / GPIO 4) on the ESP32 to facilitate serial data transmission.

Step 3: Integrating the Soil Moisture Sensor Module

- Plug the probe of the Soil Moisture Sensor into the sensor's amplifier/converter board using the provided 2-pin jumper wires.
- Connect the VCC and GND pins of the converter board to the respective power and ground rails.
- Connect the Analog Output (AO) pin of the sensor module to an Analog-to-Digital Converter (ADC) pin on the ESP32 (e.g., GPIO 34 / VP) to read the continuous analog voltage representing soil dryness.

Step 4: Connecting the 5V 1-Channel Relay Module

- Connect the VCC and GND control pins of the Relay Module to the ESP32's 5V and GND rails to power its internal optocoupler and electromagnetic coil.
- Connect the IN (Signal) pin of the relay to a Digital Output pin (e.g., GPIO 12 / GPIO 2) on the ESP32, which will handle the high/low switching logic for the pump activation.

Step 5: Wiring the 5V Mini Water Pump and Power Supply

- Cut and strip the positive wire of the external DC power source or the 5V external line.
- Route the power wire through the high-voltage side of the relay by connecting it to the Normally Open (NO) terminal.
- Connect the Common (COM) terminal of the relay directly to the positive terminal (red wire) of the 5V Mini Water Pump.
- Connect the negative terminal (black wire) of the water pump back to the common GND rail to complete the electrical circuit.

Methodology

- Data Collection: reads analog data from the soil moisture sensor and digital data from the DHT11 sensor every 5 minutes.
- Decision Making:
 - The system checks whether soil moisture falls below the dry threshold.

- The system also evaluates temperature and humidity levels to estimate potential environmental risks for plant health.

- **Normal Condition (Low Risk)**

When the temperature is between 18°C and 30°C (DHT11 reading: 18-30°C), the air humidity is between 40% and 70% RH (DHT11 reading: 40-70%), and the soil moisture is between 30% and 60% (approximately 1800-3000 ADC units), the plant is considered to be growing under healthy conditions with a low risk of disease or pest infestation.

- **Mold Growth Risk (Medium Risk)**

When the temperature is between 20°C and 30°C (DHT11 reading: 20-30°C), the air humidity exceeds 80% RH (DHT11 reading: >80%), and the soil moisture is above 70% (approximately below 1500 ADC units), the environment becomes favorable for mold development.

- **Root Rot and Fungal Disease Risk (High Risk)**

When the temperature is between 22°C and 30°C (DHT11 reading: 22-30°C), the air humidity exceeds 85% RH (DHT11 reading: >85%), and the soil moisture remains above 80% (approximately below 1200 ADC units) for an extended period, the plant faces a high risk of root rot and fungal infections.

- Pest Infestation Risk (Moderate to High Risk)

When the temperature ranges from 25°C to 32°C (DHT11 reading: 25-32°C), the air humidity is between 70% and 90% RH (DHT11 reading: 70-90%), and the soil moisture exceeds 75% (approximately below 1300 ADC units), the conditions become suitable for pests such as mealybugs and fungus gnats.

- Warning messages can be generated when conditions become favorable for fungal diseases, mold growth, or pest development.

- Execution: The relay activates the mini water pump to water the plant until the optimal moisture level is reached.

Software Implementation

- Arduino IDE is used for programming the ESP32 using C++ language.
- The ESP32 continuously reads data from the Soil Moisture Sensor and DHT11 sensor.
- Sensor values are processed every 5 minutes.
- The ESP32 connects to a Wi-Fi network and uploads sensor data to an IoT cloud platform such as ThingSpeak or Blynk.
- Users can remotely monitor temperature, humidity, soil moisture, and environmental risk conditions through an Internet-connected dashboard.

Model Performance & Results

- The system successfully monitors room temperature, humidity, and soil moisture in real time.
- The mini water pump automatically activates whenever the soil becomes too dry.
- Pulse-watering logic helps reduce the risk of root rot and soil waterlogging.
- The prototype is expected to maintain healthy plant conditions during long weekends or short vacations.
- The system can provide environmental risk warnings when temperature and humidity conditions become favorable for plant diseases or pest development.
- Environmental data can be monitored remotely through an IoT dashboard.
- Historical sensor data can be stored and visualized using cloud-based monitoring platforms.

- To establish an accurate baseline for the automated watering threshold logic, an experiment was conducted on June 21, 2026, using completely dry soil before any irrigation took place. The real-time logging data captured from the sensors is structured into the evaluation picture below:

created_at	entry_id	Temperature	Humidity	Soil Moisture
2026-06-21	17	33.9	67.1	4095
2026-06-21	18	33.9	66.4	4095
2026-06-21	19	33.9	65.9	4095
2026-06-21	20	33.8	65.3	4095
2026-06-21	21	33.7	65.6	4095
2026-06-21	22	33.8	66.9	4095
2026-06-21	23	33.8	67.5	4095
2026-06-21	26	33.9	67.4	4095
2026-06-21	27	33.9	67.7	4095
2026-06-21	28	33.9	68.7	4095
2026-06-21	29	33.9	69.1	4095
2026-06-21	30	33.8	68.9	4095
2026-06-21	31	33.7	68.6	4095
2026-06-21	32	33.7	68.1	4095

- Under these extreme dry readings ($ADC = 4095 > 3500$), the system successfully validated its automation threshold logic, instantly flagging a high-priority irrigation request to trigger the 5V relay module and activate the mini water pump.

created_at	entry_id	Temperature	Humidity	Soil Moisture
2026-06-21	61	31	80	2201
2026-06-21	62	31.3	78.3	2271
2026-06-21	63	31.3	78.3	2269
2026-06-21	64	31.3	78.3	2276
2026-06-21	65	31.3	78.2	2270
2026-06-21	66	31.3	78.2	2182
2026-06-21	67	31.3	78.1	2423
2026-06-21	68	31.2	78.2	2566
2026-06-21	69	31.2	78.2	2572
2026-06-21	70	31.2	78.2	2569
2026-06-21	71	31.3	78.3	2581
2026-06-21	72	31.3	78.4	2577
2026-06-21	73	31.3	78.8	2378
2026-06-21	74	31.4	78.8	2495
2026-06-21	75	31.4	78.6	2539
2026-06-21	76	31.4	78.5	2579

- As detailed in the before picture, the activation of the 5V mini water pump yielded an instantaneous and measurable response in the soil's physical properties. The analog raw readings from the soil moisture sensor dramatically dropped from the saturation limit of 4095 down to a stabilized operating envelope between 2182 and 2581. The continuous data stream confirms that the cloud-connected IoT(ThingSpeak) database architecture functions reliably in recording consecutive entry logs seamlessly over Wi-Fi.

System Testing

- Verify all hardware components are connected correctly.
- Power on the ESP32 and sensors.
- Observe sensor readings through the Serial Monitor.
- Dry the soil slightly and confirm that the relay activates the water pump.
- Check whether the pump stops automatically after the optimal moisture level is reached.
- Run the system continuously to evaluate stability and reliability.
- Verify Wi-Fi connectivity and successful cloud data transmission.
- Confirm that sensor readings are updated correctly on the IoT dashboard.

Troubleshooting

Soil Moisture Sensor Not Responding

- Check wiring connections and verify the analog pin configuration.

Pump Not Working

- Ensure the relay module receives sufficient power and confirm correct relay wiring.

Incorrect Sensor Readings

- Recalibrate the sensor threshold values and inspect sensor placement.

Arduino Not Uploading Code

- Verify USB cable connection, COM port selection, and Arduino IDE configuration.

Internet Connection

- Verify Internet connectivity, API keys, and dashboard configuration.
- Wifi connection failure.

Future Development

Integration of Machine Learning Models: Future iterations could implement lightweight Edge AI or tinyML models directly on the ESP32. By analyzing historical data of soil moisture degradation alongside localized weather forecasts, the system could predict exactly when a plant will require water rather than relying on static threshold logic.

Custom Mobile Application: Developing a dedicated, native cross-platform mobile application (using Flutter or React Native) to replace third-party dashboards like Blynk or

ThingSpeak would provide a more tailored user experience, featuring customizable plant profiles and localized push notifications for disease warnings.

Scaling up to Home Vegetable Garden (Family Garden) Automation: The system architecture can be upgraded from a single indoor pot monitor to a multi-channel controller capable of managing a backyard or balcony family vegetable garden. The system can be expanded to manage specialized domestic gardening structures such as multi-tier vertical vegetable racks, hydroponic PVC pipe setups, or grouped Styrofoam planting boxes . Instead of a single water pump, a sub-irrigation network consisting of low-pressure drip irrigation tapes or micro-sprinklers can be installed to distribute water evenly across different vegetable beds (e.g., leafy greens, herbs, and root vegetables)

Conclusion

The Smart Plant Monitor and Automated Watering System provides an effective and affordable solution for maintaining indoor plants automatically. By integrating environmental sensors, ESP32-based control logic, and an automated watering mechanism, the system reduces manual effort and ensures healthier plant growth. In addition to automated irrigation, the system can be extended to monitor environmental conditions and provide early warnings about potential disease or pest risks. This demonstrates how low-cost IoT-based smart plant monitoring and automated irrigation system can improve plant care, convenience, and preventive plant health management. By integrating environmental sensors, ESP32-based control logic, automated irrigation, and cloud connectivity, the system enables real-time monitoring and remote access to plant health information. The project demonstrates how low-cost IoT technology can improve plant care, convenience, and preventive plant health management.

APPENDIX

<https://github.com/LeTrungDuc10423193/smart-plant-monitor-and-automated-watering-system>